

LSAT Experiment

About the experiment:

- 4 groups:
 - A: prediction only
 - B: prediction + LLM explanations
 - C: prediction + Experts explanations
 - D: prediction + LLM confidence
- LLM used: Chat Gpt 5
- 10 LSAT questions in the experiment
 - 6, where our LLM model was correct, and also the paper model
 - 4, where our LLM model was wrong and the paper model was wrong in his first or second choice → or in simple words: 4 questions when LLM was wrong
- Accuracy:
 - The paper model: $13/20 = 65\%$
 - the LLM: $16/20 = 80\%$
 - The experiment system accuracy = $6/10 = 60\%$

Results for 20 participants in each group:

----- Accuracy Table: User Choice vs. True Choice -----

	Group	Correct answers	Total questions	Accuracy	Participants	Correctly Predicted Answers	Incorrectly Predicted Answers
0	A - prediction only	97	200	48.5%	20	75/120 ~ 62.5%	22/80 ~ 27.5%
1	B - prediction & LLM	145	200	72.5%	20	107/120 ~ 89.2%	38/80 ~ 47.5%
2	C - prediction & expets explanation	111	200	55.5%	20	82/120 ~ 68.3%	29/80 ~ 36.2%
3	D - prediction & confidence	94	200	47.0%	20	66/120 ~ 55.0%	28/80 ~ 35.0%

results:

1. Chi-Square for comparing all groups:
 $\chi^2 = 33.23$, $p = 2.87 \times 10^{-7} < 0.001$ → there is a statistical significant difference between all groups
2. Z pairwise test:

groups	z	p	statistically significant?
A vs B	-4.91	9.132078384593694e-07 <u>$p < 0.001$</u>	Yes
A vs C	-1.40	0.1611777894498059 9	No
A vs D	0.30	0.7639451206373693	No
B vs C	3.54	0.0003976075822300418 <u>$p < 0.001$</u>	Yes

B vs D	5.20	1.994819587432688e-07 <u>p < 0.001</u>	Yes
C vs D	1.70	0.08903099666390431	No

→ B is statistically significant